



Solar pumps can bring great benefits (Credit: govinfo.me)

water, which is a major modern-day environmental concern.

Flybird Innovations, for instance, provides a solution called Siri Automatic Irrigation Controller, which helps in sensor-driven automated irrigation and fertiliser distribution processes.

Sensorised fertigation

Fertilisers and chemicals like pesticides and weedicides are essential components for productive farming. Like water, these too need to be applied at an optimally accurate amount—under-use will not deliver their purpose, while over-use might harm the lands.

Sensors come into play for automating this process as well. Different sensors specific to different chemicals measure the amount of the corresponding chemicals present in the soil. In addition, sensors measuring different correlated parameters like pH levels and chemical composition of soil are used for enhancing measurement accuracy. Depending on all readings, a microprocessor unit runs fertiliser and chemical sprays till the desired amount is distributed.

Remote control and auto-scheduling

Innovations to auto-schedule farm equipment—be it machines or irri-

gation controllers—can greatly help farm owners. Satish points out, “Due to erratic agriculture power supply, a farmer needs to be present at the farm at odd times, including at midnight, to irrigate the farm.” It becomes even more difficult for those whose farm yards are at a distance from their residence. Water supply availability is also timebound. Therefore devices that can automate pumps, irrigation pipes or other machines can come in handy.

One of the latest products from Energy Bots, a smart irrigation controller, helps the user to turn the taps on or off through basic mobile services. Bhatt explains, “It is a GSM device with a pre-activated SIM card inside. Once installed at site, the user can simply give a missed call from the phone on this SIM card, and that will turn the device on. The same process can again turn it off. This can also be done through SMS. Moreover, through SMS, the farmer can also check the running status of the device. We are also working on a software platform using which pre-scheduling and data analytics can be availed.”

Another startup called Avanjilal Agri Automation Pvt Ltd provides a mobile application-driven irrigation system. Farm owners can schedule their systems through their smartphones based on time. Apart from that,

sensors placed in the field enable timely irrigation as part of this solution.

Many such solutions by different vendors throughout India are bringing automation closer to the farms rapidly.

Solar setups

Solar panels combined with automated instruments is a big boon for the agri community. Solar pumps are becoming increasingly popular for the advantages these bring. Areas with limited-to-no electricity can ensure storage of water by running the pumps using trapped solar energy.

S.S. Pradhan, general manager - sales, Rotomag Motors and Controls, says, “India gets sunlight 305 days a year. So, running solar pumps throughout these days is simple. Sixty days are mostly rainy, while the rest are dry days. This can be easily managed with stored water.”

With water available throughout the year, irrigation and farming will remain consistent and the yield will keep coming. Moreover, solar pumps provide a big environmental relief compared to diesel pumps, which emit toxic gases and other pollutants.

Implementations in India

Smart farming has seen impressive adaptation in India with many success stories. Satish gives the example of Flybird's recent installation where farm owner Chandu Gavit from Bokalzar village, Maharashtra, installed Siri Irrigation System. Gavit testifies, “Initially, there was the challenge of travelling to the farm during odd hours to get the pump started or irrigation lines running. Implementation of automated irrigation lines has solved these challenges.”

Not only have the daily challenges been addressed, automation has also cut down some major expenses for the owner. Gavit explains, “Labour cost for this kind of work is huge. Implementing these technologies has enabled us to cut that cost down to almost zero.” The biggest benefit came in the form of improved yield. “Crop production improved by up to 15 per cent, compared to what was being produced manually,” confirms Gavit.

Karnataka launched the biggest drip irrigation project at Ramthall region earlier this year. To address the incom-

